

AI DISCLOSURE

Manuscript: Institutional Observability Under Scaled AI Governance: Deployment-Scale Capacity Degradation in Human Oversight Pipelines

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Declaration of Generative AI and AI-assisted technologies in the writing process

This statement appears in the manuscript immediately above the references, per Elsevier policy.

During the preparation of this work, the author used **Windsurf (Cascade AI)** in order to assist with:

1. **Manuscript formatting and compliance checking** — ensuring the manuscript meets Technology in Society and Elsevier submission requirements, including structured abstract formatting, reference style conversion, and declaration preparation.
2. **Submission package preparation** — generating cover letter, highlights, author declarations, ethics statement, data availability statement, and other required submission artifacts.
3. **Language editing and clarity review** — improving readability and consistency of the manuscript text.

After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Core Intellectual Contributions

The following elements were developed independently by the author without AI assistance:

- The state-space model of oversight saturation dynamics
- The SEDI (State-Estimation Degradation Index) formulation
- The Kingman heavy-traffic approximation analysis
- The three governance-by-design principles
- The identification and analysis of the three compounding mechanisms (assurance surface inflation, accountability routing fragmentation, review-depth asymmetry)
- All mathematical derivations and proofs
- The Monte Carlo simulation design and parameterization

Tools Used in Research Process

- **Simulation code:** Python with NumPy, SciPy, Matplotlib (standard scientific computing libraries; no generative AI involved)
- **Reference management:** Manual verification of all citations; no AI-assisted reference generation
- **Figure generation:** Matplotlib and manual layout; no AI-generated images

Compliance with Elsevier Policy

- No AI tool is listed as an author or co-author.
- All AI-generated output was reviewed and verified for accuracy.
- The author maintains full accountability for the work's accuracy and integrity.
- Grammar and spell checking tools were used without requiring disclosure.

Statement prepared by: Shaurya Gupta **Date:** June 3, 2026